EXPERIMENTAL ELECTROACOUSTIC FEEDBACK SYSTEMS, THE PERFORMER AND THEIR AUDIENCE

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ABSTRACT

The paper presents a phenomenological description of an experimental approach to solo improvised electroacoustic performance utilizing acoustic space, a Logic Pro environment buss array, Max/MSP and inherently unstable acoustic, electronic audio and midi feedback loops. Thematic dissolution is the musical theme following notions of indeterminacy in the composition and performance of musically organized sound. Embodied, performative engagement within acoustic space, both conceptual and practical, passive and active, is discussed in relation to experimental composition, the technological setup, the sounds themselves and the audience.

1. THEORETICAL BACKGROUND

Experimental approaches to electronic music composition and performance are as diverse as the composers who have taken on experimentalism as a possibility for their music practice[1] [22, 23]. This paper does not attempt to posit any universal definition nor genre for the term 'experimental music' but rather sets out to describe one possible method, its technological setup, compositional strategies, and the performative dynamics involved in such an approach.

This particular experimental approach has evolved from a number of phenomenologically grounded research interests centred around Pierre Schaeffer's [30] notion of acousmatic listening and electronic transformation of soundscapes; John Cage's notions of indeterminacy in composition and performance [5, 6]; Alvin Lucier's[2] use of whole acoustic spaces in music performance; and, following Pauline Oliveros [24], electroacoustic improvisation as a guided attentional process. Its medium is non-tonal, timbrally focused musically organised sound.

Phenomenological inquiry in this context, following the early Husserl and Heidegger[3], is a 'how to' rather than a 'what' - it requires a methodologically based description of 'one's own' lived experience, or musical experience in this case. An inductive rather than deductive technique, this phenomenological description attempts to open up musical possibilities by disclosing the musical phenomena themselves as they arise in the process of performance. As such, this theoretical focus is derived not from a theory about music but from a music praxis itself.

From this perspective, the analytic context for a descriptive analysis of musical performance revolves around uncovering the constantly changing, meaningful relations involved in the solo electroacoustic improvisatory practice itself. That is, the relational (intentional) processes of engagement between the performer and:

- their technical setup
- the graphic score
- the acoustic space
- the audience
- the sounds themselves

It should be noted that attentional processes are one form of intentional relation, intentionality however also encompasses meaningful relations that are not necessarily explicitly disclosed in one's attention. For example, while attending to a graphic score during performance one cannot also at the same time focus attention on the acoustic space, and yet the acoustic space remains a meaningful (intentional) context within which the performance takes place. Furthermore, practical understanding - or an embodied relation to these various aspects of musical performance - rather than rational thought processes, is prioritized in relation to the ongoing auditory perception (aesthesis) of the sounds themselves. Music in terms of both performance and listening, is here assumed to be an embodied, ongoing, temporalizing process of meaningful relations in constant flux[4].

[1]Both Pierre Schaeffer's phenomenologically influenced acousmatic approach to experimental music and John Cage's experimental notions of indeterminacy are, however, central to this experimental theme.

[2]Alvin Lucier's use of acoustic space in compositions such as *I am Sitting in a Room* and *Shelter* [19] along with my participation in the 2007 Transnational Ecologies internet performance of *Quasimodo the Great Lover* [28] and Lucier's notions of acoustic exploration [20] have particularly informed my understanding of the compositional possibilities involved here.

[3]My Doctoral specialty [26] is in the convergence and divergence of phenomenological concepts in the early works of Husserl and Heidegger. This descriptive approach is largely derived from Husserl's Logical Investigations [13] through to Ideas Book 2 [14] as well as Heideggers early transcendental phenomenology [10, 11].

[4]A more detailed discussion of this phenomenological approach is included in a paper for the forthcoming THNMF Conference 2009

2. THE PROBLEMATIC

2.1. Disclosing Performative Relations

The electroacoustic performance setup described below (see 3.1 Experimental Setup) has evolved out of a practical exploration of the composition and performance relations involved in solo electroacoustic improvisation. The experimental aim is to uncover in a practical sense those relations as they arise in the performance itself in order to then describe the phenomena and their dynamics. The compositional approach thus uses Oliveros' notion of improvisation as an attentional process [24, 31] to set up performance relations that specifically require attention to the reverberant acoustic space within which all performance and audience takes place.

The electroacoustic setup (any transducer such as a dynamic microphone) monitors the acoustic space setting up a feedback loop[5] between transducer and speakers, with electronic treatment of the sound transforming the natural soundscape of the performance space. From the performer's perspective this setup requires focused attention on the acoustic properties of that space as they transform the audio input/output over the duration of the performance. Interestingly this also includes the acoustic damping effect audience members' physical bodies have on that space as well as any ambient noise emanating from the audience itself.

In terms of the attention paid to musical structure, improvisation does not require following a more or less strictly defined set of tonal and rhythmic instructions. On the contrary it requires an openness to the sounds themselves as they occur [2] and this openness is reinforced by a composition structure that, following Cage [6], is indeterminate with respect to performance.

An acousmatic approach to the performance complements the indeterminacy of the improvisation by emphasizing the performative attention to the sounds themselves as they arise and without reference to their originating source. Acousmatic transformation of the soundscape[6] is accomplished electronically with attention to the timbral character of the mix as it evolves. This forward looking musical attention is not concerned with developing a pre-conceived theme but instead responds to the indeterminate immediacy of the initial soundscape shaped by the acoustic space. Subsequent electroacoustic inputs monitor this transformation via the speaker to electroacoustic input feedback loop and the development of the musical theme as a whole evolves out of this unpredictable and circular signal path. The indeterminate focus here is on thematic dissolution as a musical theme in itself.

2.2. Aesthetics as Aesthesis

While many experimental approaches to music have been criticized for their 'sterility' or lack of musicality, and this is perhaps especially the case with Cage's indeterminate approach to performance [18], it remains that the musical/noise outcome must primarily be a product of the compositional concepts at work rather than the composer's or performer's aesthetic choices. However, it does seem rather artificial to exclude aesthetics from a musical performance that is composed to uncover a performer's dynamic relation to music.

The performer's relation to the sounds themselves is in phenomenological terms fundamentally a process of aesthesis or straightforward sensory perception that is always accompanied by a flux of affect or feeling (pleasant, unpleasant and neutral). Rather than abrogating aesthetic choice entirely the composition structure requires a degree of attention towards the timbral qualities of the sounds themselves in order to uncover this important musical relation. The performer thus improvises the electronic mix by following what sounds 'pleasant' or 'unpleasant' within the constantly evolving feedback loop.

A focus on timbre and aesthetics also brings to light the improviser's problematic relation to their audience [1]. Focusing on the audience members' possible reception and aesthetic valuation of the performance can be a serious distraction for a performer, and improvisers have devised various means to neutralize this tendency[7]. For this experimental composition, the acousmatic transformation of the acoustic space's soundscape should be guided by the performer's aesthetic choices in response to the indeterminacy of the constantly evolving sounds. The use of the acoustic space itself also bodily involves the audience and any sounds they might make thus both problematizing and making use of the performer's dynamic relation to their listeners.

3. METHODOLOGY

3.1. Experimental Setup

A transducer (condenser, dynamic or contact microphone, electric guitar/bass and so on) monitors the acoustic space and provides the only audio signal input into a laptop computer running Apple's Logic Pro mixing environment controlled by a hardware mixer. The

Proceedings [27]. While my primary phenomenological sources are Husserl and Heidegger the relation to music has also been informed by Lochhead [18] and Clifton [7, 8] amongst other phenomenologically influenced musicological approaches [3].

[5]Electroacoustic compositions using various forms of feedback began in the 1960's, David Tudor's 1969 *Rainforest* work being an excellent example of electronic feedback systems [29]. While the Larsen effect of runaway acoustic feedback (squealing microphones) does play a part in the current composition under discussion here, it is merely one form of feedback. I am also interested in feedback systems as a whole, including electronic, digital data, logical and practical/performative feedback and their relation to composition [4, 21].

[6]There is an interesting convergence of soundscape and acousmatic methods involved in monitoring an acoustic space during performance. The initial audio input is the soundscape of the acoustic space whereas subsequent electronic manipulation of the sound transforms the soundscape, obscuring its origins. The acousmatic output via the speakers then becomes the electroacoustic feedback input and so on.

[7]Charlie Parker for instance would often play with his back to the audience in order to concentrate on the improvisation [1].

raw mono audio input is bussed to any of six auxiliary channels, each with a set of EQ, compression and effects inserts. Each auxiliary can be bussed to any of the five others resulting in a 6x6 buss matrix (see Figure 1). In terms of the possible signal path bus combinations, this simple auxiliary array offers 2^{30} or around 1 billion choices for routing the digital audio signal internally[8].

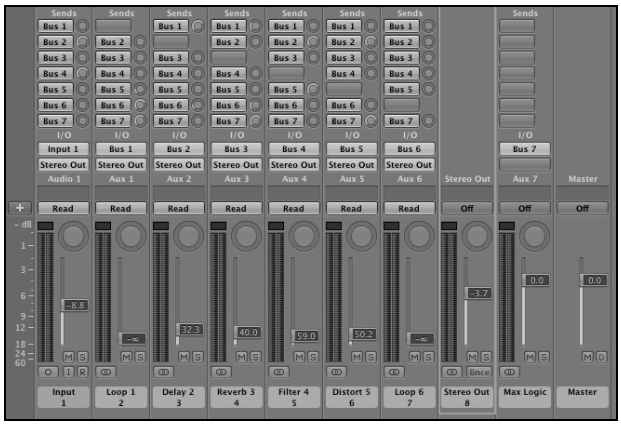


Figure 1. A simple 6x6 auxiliary buss array in Logic Pro offers 2^{30} possible combinations for mixing.

Two audio channels mixed from the input channel or any auxiliary are bussed to a Max/MSP patch for sinusoidal analysis. The raw waveform data outputs are scaled and variable speed limited to produce 24 independent MIDI controllers that are mapped back onto various insert parameters across the Logic auxiliary channels (see Figure 2). Dynamic insert parameter control is thus automated by the audio signal itself.

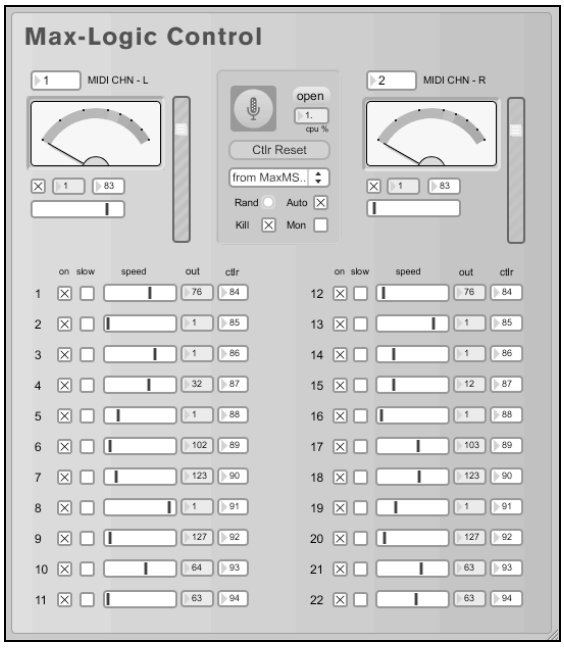


Figure 2. Max/MSP sinusoidal analysis to MIDI controller patch.

[8]Each auxiliary channel has 5 busses. Assuming each buss is either on or off there are 2^5 or 32 possible combinations per auxiliary, or 2^{30} = 1073741824 routing paths across all 6 channels.

Sidechain compression is used across all auxiliaries to help automate the mix. The improvisation largely consists of riding the faders and EQ settings, and navigating through the buss structure, setting up signal path loops and controlling acoustic and electronic feedback.

The technical setup is arranged for maximum sensitivity to the acoustic space, transforming the electroacoustic input that in turn drives the parameter control and provides the output and raw audio input signal for ongoing improvised mixing choices in response to that reverberant acoustic space. Complexity in the auxiliary buss array allows a high degree of freedom in the improvisation and provides the technological potential for navigating between electroacoustic serendipity and chaos.

The live performance audio input is recorded at 24bit 48kHz (effectively a field recording) along with all automation in Logic and provides the raw material for further studio editing and mastering.

3.2. Composition Structure

The electroacoustic technology and its signal paths define the limits and potential of the compositional structure. Use of a graphic score (see Figure 3) and performance instructions showing possible buss patterns and other interactions over time allows for a structural approach to the improvisation. The specific properties of any acoustic space (including microphone and speaker placement within it, audience interaction and so on), uniquely define the acoustic response of the feedback system as a whole such that no two performances of the identical score and technical setup can be the same.

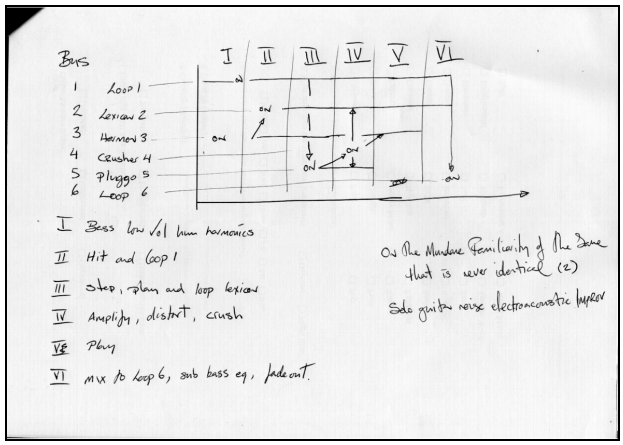


Figure 3. Early draft of graphic score and instruction set for June 2009 performance.

Likewise, the fragile complexity of the auxiliary buss connections, vulnerable to escalating electronic feedback that then drives electroacoustic feedback and dynamically changes each channel's insert parameters, is akin to wandering through a compositional forest of one billion trees at night without a compass in a storm[9]. This

[9]After Heidegger [12], *Holzwege*, or forest paths, is a metaphor for a phenomenological method of inquiry. Essentially exploratory it rejects

compositional indeterminacy is explicitly built into the technical setup to accentuate performative uncertainty. Whilst various initial buss and insert conditions can be tested in the studio the resultant graphic score acts merely as a general guide to possible pathways through an otherwise constantly evolving soundscape and its electronic acousmatic transformation.

4. PERFORMANCE RESULTS

The composition and technical setup described above has been developed over a year's practical research. Each performance and studio experiment has developed not only the electronic techniques for the acousmatic transformation of the soundscape within an acoustic space but also the attentional awareness of the performative relations that the composition is intended to bring into play[10]. This ongoing performative development, an experimental form of practical music research, has in turn aided in the further development of my theoretical interests in a phenomenological investigation of musically organized sound.

Together, theoria and praxis are both necessary elements of phenomenological analysis where the phenomena in question can only be disclosed firstly via demonstration and then description. The experimental performance approach described in this paper is just such a practical demonstration of an ongoing music research project. The following sections are preliminary reflections on the performance results so far.

4.1. The Technological Setup

The electronic music technology used forms the practical background context for the electroacoustic performance, and an attention towards the embodied, tactile nature of performance[11] within an acoustic space has highlighted several spatial relations in that performativity.

Electroacoustic inputs via a contact, condensor or dynamic microphone, or an electric guitar/bass resonating in the acoustic space provide a direct, performative connection with that space. It is this basic electroacoustic feedback loop between the transducer input and amplified output via the whole acoustic space that provides the context for all other interactions with the audience, the electronic setup and the sounds themselves. This holistic attention to the spatial context within which the performance takes place constitutes the basis of my compositional approach[12] in that all other attentional relations are disclosed in relation to that acoustic space as a whole.

This attention to embodiment and space has emphasized a commonly reported technical problem with laptop computer performance: The use of a mouse pointer in a graphic user interface is very limiting and the inclusion of a hardware interface (in this case a Euphonix MC Mix) has greatly expanded the potential for control of both the electroacoustic and internal electronic feedback loops while allowing for more expressive treatment of the timbral qualities of the sound.

Navigating the send busses and volume faders, insert parameters such as EQ notches, compression gain and ratio, and various static effect as well as Max/MSP settings allows for a diverse range of improvisational choices throughout the performance. Due to the unpredictable, chaotic nature of the positive and negative feedback loops generated, these choices are generally driven in response to the sounds produced by the technical setup rather than a preconceived aesthetic. Any thematic development, such as resetting the busses to the next phase in the graphic score, is constantly destroyed by this inherent indeterminacy. Nonetheless - and in common with much experimental, improvised electroacoustic music as well as arguably athematic serialism - constant thematic dissolution does produce a recognizable musical form often experienced as noise, depending on one's aesthetics. Thematic dissolution is in this sense both a musical theme and an indeterminate composition concern built into the technical setup itself. It demands an openness to both the sounds themselves and the improvisation in response to those sounds, and emphasizes the dynamic temporal processes involved in the performance as a whole.

Use of Max/MSP to automate the control of insert parameters in response to the changing electroacoustic waveform adds a strong element of indeterminacy and constant change to the evolution of the sound and also helps break up feedback loops. The overall MIDI output is noticeably sensitive to the amplitude of both input audio channels into Max/MSP, each of which can be manually adjusted via the Logic busses and in the Max patch itself. Making use of three independent sinusoidal analysis patches to transform digital audio data into 24 unique waveform data streams allows for a significant degree of complexity in the Logic parameter automation. Multiple parameters can be controlled by one MIDI controller, including any insert, panning/spatialization, send buss level and so on. However, balancing CPU usage and MIDI buffer overruns with the controller data rate defines the technical limit of that complexity.

The rate of MIDI controller data sent to Logic Pro from each of the 24 controllers can be individually varied

the notion of any absolute ground or starting point for an analysis of the meaningful relations of lived experience, in this case the performative relations of electroacoustic improvisation. One chooses a path and hears where it leads.

[10]A set of performance (field) recordings *(Variations on Acoustic Feedback)* are due for mastering and release on the Slow Release Music Label in late 2010 (http://slowrelease.waapamusic.com). See also http://malcolmriddoch.com for streaming audio examples.

[11]Embodiment and spatiality are a common phenomenological theme. For a discussion of embodiment from a musical perspective see for example Corness [9].

[12]Along with Lucier the site specific, environmental works of Australian acoustic composer Alan Lamb [16] have also influenced my approach with their sensitivity to natural and human soundscapes, and auditory evocation of the spatial and temporal dimensions of place.

from 5 messages per second to one every 3 minutes. At fast rates the parameter changes can result in very densely textured audio output while slower rates tend to produce more stable responses punctuated by sudden changes. In the initial default state all MIDI controllers are on and set at the same (fast) rate producing a synchronized effect in the parameter automation and resultant audio. The rate can be randomized for each controller and automatically reset at random intervals resulting in asynchronous changes to the insert parameterization and a perceptibly more chaotic audio response that in turn produces a wider range of dynamics in the MIDI controller outputs (see Figure 4).

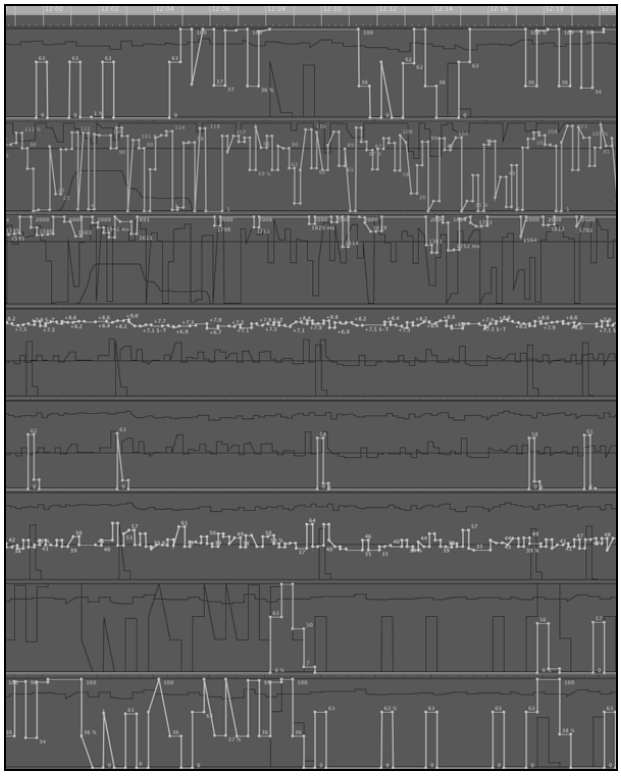


Figure 4. Photoshop composite of eight Logic Pro auxiliary tracks showing Max/MSP MIDI controller automation data over approximately 20 seconds. Recorded live at Kurongkurl Katitjin Gallery, Edith Cowan University, 21 May 2010.

The nonlinear dynamics of these various feedback systems - the basic electroacoustic loop, manifold electronic buss loops and 24 Max/MSP audio to MIDI loops - all combine and interact to produce an unpredictably chaotic yet technologically structured and thus musically organized outcome. The positive and negative feedback loops involved in this compositional setup typically produce a-periodic oscillations, abrupt changes from one more or less stable state to another, stable and unstable chaotic oscillations accompanied by sudden transient bifurcations and runaway self-reinforcing crescendos terminated by limit cycles or parameter changes. This technologically constrained, cyclically chaotic feedback system gives form to the auditory context within which all improvisation and attempts at compositional control take place.

4.2. The Graphic Score

As a set of mnemonics for operating the technological setup the graphic score aids in setting up the initial buss conditions for each phase of the performance. However, due to the dynamically unpredictable nature of any particular acoustic space and the feedback sensitivity of the electronics, while these initial buss states can be explored and noted down beforehand in the studio they can only serve as a guide during the performance itself. The auditory outcome can be radically different for the same buss structure where small differences in parameter controls can propagate rapidly through the various feedback loops.

4.3. The Audience

While an audience can often be a necessary distraction for improvisation, I have found that making the audience an object of the composition by including their interactions within the acoustic space adds a useful dimension to the performance dynamic. For example, the performance problems associated with anticipating an audience's aesthetic response to the improvised music, often viewed as a distraction by improvisers, seem to be attenuated. Instead of ignoring or shutting out the audience they become an integral part of the whole musical process. The acoustic space along with everything and everyone in it has become the musical instrument. Microphone placement can help make this relation explicit to the audience members themselves such as for instance aiming a shotgun microphone at an auditorium's acoustic 'sweet spot', quite often a few rows back from the centre, front seats.

The use of a 'black box' laptop computer can present problems for some audience members due to the relative lack of performative feedback. While a hardware interface for controlling a computer allows for a much more tactile approach to the performance it also aids in connecting the performance actions with musical results, an important aspect of audience engagement with and appreciation of computer music performance[13]. Likewise, the projection of the graphic score acts as an extension of this performative communication and assists in audience engagement with the work.

While passive audiences have an effect on the acoustic properties of the performance space simply by virtue of their body mass, thus altering the electroacoustic transducer input, they can also produce sounds in response to the performance (coughing, shuffling of feet and so on). With this in mind, and contrary to usual concert protocol, audience engagement is sometimes encouraged in the sense of inviting any sonic interventions audience members might come up with in response to their sonic environment. This adds an essentially indeterminate, discontinuous audio input into the electroacoustic and MIDI feedback loops that assists in breaking up runaway electronic and acoustic feedback

[13]For a discussion of this problem and possible solutions see for example Garth Paine on standardizing interface guidelines [25].

as well as contributing to the timbral complexity of the mix.

Giving a compositional context to these performative relations to the audience, as opposed to simply ignoring them while performing, does seem to accentuate these positive aspects for performance. The audience now has a use value for the composition and performance of the musical work itself beyond being there to merely receive, listen and critique.

4.4. The Sounds Themselves

Musically organised sound can be fascinating, both for audiences and performers alike. Forgetting oneself and becoming absorbed in the sounds themselves is perhaps one of the main pleasures and primary motivations for listening to music. I have always been fascinated by the collective fascination and focussed attention displayed by audiences listening to and watching musical performance.

Musical fascination is posited here as a mode of aesthesis in the sense of an absorption in the ongoing auditory perception of the sounds themselves as they arise and fall away. Being captivated 'in the moment', for performers and audience alike, could be said to be one general indicator of a successful performance. From the perspective of the indeterminate electroacoustic composition under discussion, aesthetic fascination can take the form of an absorption in timbral affect - the pleasant surprise associated with serendipitous mixing choices coinciding with the Max/MSP parameter automation and acoustic/electronic feedback. Following the aesthetics of the sounds themselves and allowing the acoustic space to dictate the resonances and evolution of the work as a whole comprises the overall context within which all other interactions take place.

However, for this performer, aesthetic absorption in the sounds as they are heard is an attentional process that is subject to constant dissolution due to the simple fact that the sounds themselves are just one of the relations of sense involved in the performance. One's attention also constantly deals with compositional priorities, the control of runaway feedback, and the demands of a complex and inherently fragile technical setup. I can only assume that aspects of this relational process are more or less common to all forms of music performance, acoustic and electroacoustic, determinate and indeterminate, from dot notation through to open improvisation. The musical sound provides the ongoing background context within which one deals with the composition, the instruments and the audience. Absorption in the sounds themselves remains an ongoing, open possibility.

Making this dynamic relational complex an explicit object of the composition does seem to aid in separating out the various attentional processes. While absorption in the sounds themselves also relies on becoming increasingly competent with the instrumentation and the composition, an awareness of how these processes interact could perhaps assist in being open for the moment one's attention might become absorbed.

5. CONCLUSIONS

Experimental music composition and performance as a disclosure of the sounds themselves in their lived, relational, spatial and embodied context would seem to be a complementary practical methodology for phenomenological music research. Practical awareness of the attentional processes involved in performance is enhanced by making these processes the object of the experimental composition, and the results of the performance can be used to further investigate the phenomenal relations at play.

As regards this preliminary descriptive investigation, the performance of improvised, electroacoustic music as a whole is a dynamic process of change in the constant arising and dissolution of musical attention. It is a play of attention to composition, technology and audience within the reverberant acoustic space of the sounds themselves - and all of these intentional relations are subject to constant dissolution as a flux of attentional processes for the duration of the performance.

The sounds themselves remain constant however as either the background auditory perception (also a mode of aesthesis) informing the practice or as the foreground object of aesthetically absorbed attention. In other words, one can attend to the composition, the technical setup or the audience but only within the acoustic space: The sounds themselves are the auditory context giving meaning to and ultimately guiding these performative relations. From this phenomenological perspective, the performance of musically organised sound always has an embodied, performative, lived context and the musical work as a whole is a complex network of meaningful musical relations in constant flux.

Further research into the dynamics of this relational flux is proposed in terms of the phenomenological structure of lived time or temporality (*Zeitlichkeit*[14]). This notion of time is characterized by its circular logic and is the primary logical feedback concept informing my experimental music approach. A compositional structure building on the current paper's provisional investigation into the intentional (attentional) processes involved in musical performance would require methods for the disclosure of this circular temporal structure that links musical perception, affect and practice together. 'Openness' to the sounds themselves and 'thematic dissolution' as a compositional theme are two phenomenologically derived methodological concepts under consideration for the further development of the experimental approach outlined in this paper.

[14]Zeitlichkeit is a Husserlian term for the dynamic structure of intentionality. Also termed originary time it is a basic concept for both Husserl's phenomenology of inner time consciousness [15] as well as Heidegger's early transcendental phenomenology [10, 11].

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